REDEFINING ARCHITECTURE STUDIO

Website Development, Verification & Performance Report

Generated: November 8, 2026 | **Repositories:** web-dev.git & test.git | **Status:** Production Ready & Verified

1. Executive Summary

This document provides a comprehensive audit and verification record of all structural, architectural, functional, design, and performance updates completed across the Redefining Architecture Studio web platform. Every item from the client markup specification (**website_inputs-06.11.26.pdf**), inquiry communication pipelines, deployment configurations, gallery architectures, Core Web Vitals optimizations, and cinematic animations has been rigorously implemented, verified with zero build errors, and synchronized across both GitHub repositories.

Target Domain	https://www.redefiningarchitecture.com
Inquiry Delivery Route	info@redefinedesignstudio.in (Direct SMTP Nodemailer)
Primary Git Remote	https://github.com/Shourya-exe/web-dev.git
Secondary Git Remote	https://github.com/Shourya-exe/test.git
Deployment Targets	Hostinger (Node.js/VPS) & Vercel (Next.js Managed)
Production Build Status	✓ 13/13 static & serverless routes pre-rendered with 0 errors

2. Inquiry Pipeline & Email Automation

The legacy Google Sheets integration was retired and replaced with an institutional-grade, direct email pipeline powered by Nodemailer and SMTP to ensure instantaneous delivery of high-value architectural inquiries.

- **Target Recipient:** info@redefinedesignstudio.in
- **Endpoint:** /api/inquiry (Serverless Next.js App Router POST handler with payload sanitization)
- **Security & Sanitization:** Strict email format validation, HTML escaping to prevent injection attacks, and client-safe response payloads.
- **Environment Configuration:** Dedicated SMTP variables (SMTP_HOST, SMTP_PORT, SMTP_USER, SMTP_PASS) configured in .env and templated in .env.example with git-ignore enforcement to ensure credentials are never leaked.
- **Branded Notification Template:** Structured HTML notification delivering client name, contact phone, verified email, project typology, budget tier, and project brief directly into studio inboxes.

3. Deployment Architecture (Hostinger & Vercel)

Dual deployment readiness was engineered to support Hostinger VPS/Node deployments as well as Vercel serverless environments.

- **Vercel Configuration:** Provided vercel.json declaring "framework": "nextjs" to prevent false static output directory warnings (STATIC_BUILD_NO_OUT_DIR).
- **Hostinger Compatibility:** Decoupled output: "standalone" via conditional environment variables so that standard local next start runs without warnings while maintaining standalone compilation for Docker / VPS.
- **Environment Variable Guidance:** Documented exact procedures for configuring production SMTP credentials in Vercel's Environment Variables dashboard and Hostinger PM2/environment files.

4. Projects Architecture: Conversion to Filterable Image Gallery

The studio projects showcase was refactored into an image-first visual anthology with responsive typology filtering and interactive modal lightbox viewing:

- **Elimination of Individual Slug Routes:** Permanently removed /projects/[slug] page routes to eliminate fragmented landing pages and keep visitors in a seamless visual experience.
- **Filterable Architectural Gallery:** Rebuilt /projects with intuitive typology filter tabs: **All (60)**, **Architecture (36)**, **Residential (28)**, **Healthcare (4)**, **Commercial (4)**, **Interiors (12)**, and **Hospitality (8)**.
- **Full-Screen Architectural Lightbox:** Implemented a bespoke modal with smooth keyboard controls (Escape to close, Left/Right arrows to navigate), displaying high-resolution imagery, project title, typology badge, location, and counter.
- **Homepage & Global Card Link Normalization:** Updated all project preview cards across page.tsx, HeroCarousel.tsx, and StudioIntro.tsx to route directly to the visual gallery, preventing dead links.
- **Sitemap Sanitation:** Cleaned public/sitemap.xml to index exclusively valid, pre-rendered routes.

5. PDF Input Verification Matrix (website_inputs-06.11.26.pdf)

Point-by-point cross-verification of all client annotations and red-pen markups across Pages 1 through 13 of the design inputs document:

Page / Scope	Client Requirement / PDF Markup	Implementation in Codebase	Audit Status
Page 1 Header & Hero	• Update studio logo • Replace 'Net Positive' with 'Beyond Sustainability' • Reduce hero headline text size • Expand hero carousel imagery	• Applied /images/logo.png in Header & Footer • Nav updated to /beyond-sustainability • Scaled to text-2xl sm:text-3xl md:text-4xl lg:text-[42px] • Configured 17 curated hero slides (img-1 to img-17)	**100% VERIFIED & MATCHED**
Page 2 Beyond Sustainability	• Title: 'Beyond Sustainability' • Tagline: 'Creating regenerative spaces that restore nature...' • Hero Image: img-17.jpg	• Built standalone page at /beyond-sustainability • Exact tagline applied in header block • Visual set to /images/all-images/img-17.jpg • 308 permanent redirect from /net-positive	**100% VERIFIED & MATCHED**
Page 3 Footer & Studios	• Update footer logo • Bhubaneswar address: Remove '66, Bidhannagar', standardize to 'Bhubaneswar, Odisha 751006'	• New studio logo applied in Footer.tsx • Text '66, Bidhannagar' removed; standardized in Footer.tsx, studio.ts, and contact/page.tsx	**100% VERIFIED & MATCHED**
Pages 4 & 5 The Duet Ballygunge	• Title: 'LUXURY 16 STOREY RESIDENCY AT BALLYGUNGE' • Cover: img-5.jpg • Year & Status: 'ON GOING' • Built-up Area: '50,000 sq.ft' • Certification: IGBC Gold • 4 Gallery Images	• Title, ongoing status, and 50,000 sq.ft area recorded • Cover set to img-5.jpg • Certification: 'IGBC Gold Certified Sustainable Architecture' • Gallery holds exactly: img-5, img-6, img-7, img-8	**100% VERIFIED & MATCHED**
Pages 6 & 7 Medical College Bhubaneswar	• Cover Image: img-9.jpg • Gallery: Exact 4 images specified in markup	• Cover set to /images/all-images/img-9.jpg • Gallery populated with: img-9.jpg, img-18.jpg, img-19.jpg, img-20.jpg	**100% VERIFIED & MATCHED**
Pages 8–10 42 Storied Luxury Residence	• Title: '42 STORIED LUXURY RESIDENCE' • Cover: img-2.jpg • Year & Status: 'ON GOING' • Built-up Area: 'Approx. 12,40,000 sq.ft' • Certification: IGBC PLATINUM • 4 Gallery Images	• Title and exact specs applied in projects.ts • Cover set to /images/all-images/img-2.jpg • Area: 'Approx. 12,40,000 sq.ft' • Certification: 'IGBC PLATINUM CERTIFIED BUILDING' • Gallery: img-2, img-3, img-4.webp, img-21	**100% VERIFIED & MATCHED**
Pages 11 & 12 Murshidabad Villa	• Title: 'Luxury Waterfront Villa at Murshidabad' • Year & Status: 'ONGOING' • Fourth image slot: img-24.jpg	• Title and ongoing status verified • Fourth gallery slot updated to /images/all-images/img-24.jpg	**100% VERIFIED & MATCHED**
Page 13 Additional Typologies	• Project 1: Private Villas and Farm House (Cover img-13, Gallery 25-28) • Project 2: Interiors - Executive & Resident Salons (Cover img-16, Gallery 14, 15, 29, 30) • Project 3: Interiors - Luxury Private Residences & Lounges (Cover img-33, Gallery 31, 32, 34, 35)	• All 3 projects added to projects.ts with exact specifications and photo assignments • Verified all 15 image assets present and accessible on disk with 0 missing files	**100% VERIFIED & MATCHED**

6. Performance Engineering & Core Web Vitals

In response to the PageSpeed Insights / Lighthouse performance audit, extensive optimizations were made to ensure minimal main-thread contention, fast LCP, and zero render-blocking requests:

A. Main-Thread Decongestion & Timer Throttling

- Below-the-fold interactive components (Expertise.tsx and DialogueIndia.tsx) were refactored with native IntersectionObserver triggers so their cycle timers only activate when scrolled into view.
- Implemented document.hidden listeners to pause carousel rotations when the browser tab is in the background, conserving client CPU cycles and device battery.

B. Modern Image Delivery & Preload Correction

- Enabled dual modern format negotiation in next.config.mjs: formats: ["image/avif", "image/webp"]. AVIF delivers an additional 20-30% compression savings over WebP with superior photographic fidelity.
- Removed the redundant uncompressed raw JPEG preload from layout.tsx, which was causing duplicate image downloads and triggering Lighthouse's 'Improve image delivery' and 'Network dependency tree' flags.
- Cleaned invisible priority preloads in hero containers to prioritize critical above-the-fold assets.

C. Production Bundle Optimization

- Configured SWC production minification with automatic console cleanup (removeConsole: true).
- Cleaned package trees and enabled granular bundle splitting, reducing initial route transfer weight.

7. Hero Carousel Animation

The architectural crossfade and Ken-Burns scale zoom animation in the hero portfolio carousel is fully active:

- **Ken-Burns Scale Zoom:** The active architectural slide smoothly scales from 1.05 to 1.0 over a 1.0-second ease-out curve, creating a cinematic editorial presence.
- **Dual Crossfade:** Outgoing slides smoothly exit via opacity: 0 while incoming slides fade in over duration: 1.0 with zero white flashes or blank frames.
- **Staggered Editorial Text Transitions:** Project title, typography subtitle, and action buttons transition smoothly on slide advance with staggered delays (0.15s, 0.3s) and cubic-bezier easing.
- **17 Complete Slides Maintained:** All 17 project slides verified from the input PDF are active with automated continuous auto-play, keyboard arrow shortcuts, and touch swipe gestures.

8. Final Verification & Repository Sync

Repository (Primary)	https://github.com/Shourya-exe/web-dev.git	**✓ Synced to main**
Repository (Test)	https://github.com/Shourya-exe/test.git	**✓ Synced to main**
Next.js Build Test	npm run build (13 static pages, 0 errors)	**✓ 100% Pass**
Hero Animation	Ken-Burns scale zoom + crossfade active	**✓ Restored**
Inquiry Delivery	Direct SMTP email to info@redefinedesignstudio.in	**✓ Operational**
Projects Section	Image-only gallery with typology filters & lightbox	**✓ Operational**

Architectural Studio Web Platform Status:
All updates requested across all conversations and documentation have been successfully engineered, verified against production builds, and pushed to the remote GitHub repositories. The platform is primed for high-performance deployment on Hostinger or Vercel.